

Art Museum Data Warehouse

20 Data Analysis Questions

Model Validation Reference · Gold Layer

Model	Type	Key Columns
fct_sales	Fact	sales_key, work_id, artist_id, museum_id, size_id, sale_price, regular_price, discount_amount, discount_pct, is_in_museum
dim_artist	Dimension	artist_id, artist_name, nationality, style, birth_year, death_year, era, artist_status
dim_artwork	Dimension	work_id, artwork_name, style, subject_tags, artist_id, museum_id
dim_canvas_size	Dimension	size_id, label, width_inches, height_inches, area_sq_inches, size_category
dim_museum	Dimension	museum_id, museum_name, city, state, country, opening_days_per_week, avg_daily_open_hours, is_open_weekends

Difficulty Mid Mid-High High

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Q0 1

Average Discount % per Artist Era

Mid

Joins: fct_sales → dim_artist

Purpose

Measure whether pricing strategy (discount depth) varies across historical artist eras. Useful for commercial teams to understand which era's work gets the deepest markdowns and whether the discount structure is intentional.

Output Columns

Column	Description
era	Artist era bucket from dim_artist (e.g. Renaissance, Modern)
artist_count	Distinct artists in that era
artwork_count	Distinct work_ids sold in that era
avg_sale_price	Mean sale price across all size variants
avg_regular_price	Mean regular price
avg_discount_pct	Mean discount_pct (NULL rows excluded)
max_discount_pct	Deepest single discount in the era
pct_rows_discounted	Share of fact rows where discount > 0

Validates: era CASE logic in dim_artist, discount_pct accuracy in fct_sales

Q0 2

Canvas Size Category vs Total Revenue & Avg Discount

Mid

Joins: fct_sales → dim_canvas_size

Purpose

Identify which physical canvas sizes drive the most revenue and carry the deepest discounts. Key for inventory and pricing decisions. Also validates that size_category bucketing aligns with revenue expectations.

Output Columns

Column	Description
size_category	Small / Medium / Large / Extra Large / Unknown
sku_count	Distinct (work_id, size_id) combinations
total_sale_revenue	Sum of sale_price
total_regular_revenue	Sum of regular_price
total_discount_given	Sum of discount_amount
avg_discount_pct	Mean discount %
avg_area_sq_inches	Mean canvas area in that bucket

revenue_share_pct	Bucket's share of total sale revenue
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Validates: *size_category* bucketing, *area_sq_inches* computation, *fct_sales* → *dim_canvas_size* join

Q0
3

Museums with Most Artworks vs Operating Hours

Mid

Joins: fct_sales → dim_museum

Purpose

Correlate how many artworks a museum hosts (and their total revenue potential) with the museum's weekly operating hours. Helps spot museums punching above or below their footprint. Tests the hours enrichment logic from museum_hours.

Output Columns

Column	Description
museum_id / museum_name	Museum identifier and display name
country / city	Location fields (COALESCE'd to Unknown if null)
opening_days_per_week	From dim_museum hours aggregation
avg_daily_open_hours	Average daily open duration in hours
is_open_weekends	Boolean from dim_museum
total_artworks	Distinct work_ids in fct_sales
total_sale_revenue	Sum of sale_price
revenue_per_open_day	Total revenue ÷ opening_days_per_week

Validates: dim_museum hours enrichment CTEs, COALESCE on city/country, LEFT JOIN to hours_stats

Q0
4

Share of Artworks Priced Above Median per Nationality

High

Joins: fct_sales → dim_artist

Purpose

For each artist nationality, measure how many of their artworks have an average sale price above the overall median. Reveals premium vs discount nationalities and tests complex window/aggregate patterns including PERCENTILE_CONT and a CROSS JOIN to a scalar CTE.

Output Columns

Column	Description
nationality	From dim_artist (COALESCE'd to Unknown)
artist_count	Distinct artists in that nationality
total_works	Distinct work_ids
works_above_median	Works with avg_sale_price > global median
pct_above_median	Percentage above median
avg_sale_price	Nationality-level mean sale price

global_median_ref	The global median value (same on every row for reference)
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Validates: *PERCENTILE_CONT, CROSS JOIN scalar CTE, per-work price aggregation before nationality rollup*

Q0
5

Artworks with 'Unknown' Subject Tags That Are Discounted

Mid

Joins: fct_sales → dim_artwork

Purpose

Surface artworks whose subject metadata defaulted to 'Unknown' (COALESCE fallback in dim_artwork) but are still being discounted commercially. These represent a data-quality gap — items are active in pricing but not properly classified by subject.

Output Columns

Column	Description
work_id / artwork_name	Artwork identifier and name
style	From dim_artwork (may be NULL via NULLIF blank trimming)
subject_tags	Will be 'Unknown' for all rows in this query
size_count	Number of size variants in fct_sales
avg_sale_price / avg_regular_price	Pricing metrics
avg_discount_pct / max_discount_pct	Discount depth
is_in_museum	BOOL_OR of is_in_museum flag

Validates: COALESCE('Unknown') fallback in dim_artwork.subject_tags, discount_pct accuracy

Q0
6

Weekend-Open vs Weekday-Only Museum Pricing Comparison

Mid

Joins: fct_sales → dim_museum

Purpose

Compare average sale prices and discount depths for artworks housed in museums that open on weekends vs weekday-only museums. Weekend accessibility may correlate with broader audience and different price positioning.

Output Columns

Column	Description
weekend_status	'Weekend Open' / 'Weekday Only' / 'Unknown'
museum_count	Distinct museums in the group
artwork_count	Distinct works
sku_count	Total (work × size) rows
avg_sale_price / avg_regular_price	Price metrics
avg_discount_pct	Mean discount %
total_revenue	Sum of sale_price
avg_opening_days_per_week	Average weekly opening days

Validates: *is_open_weekends* boolean derivation in *dim_museum*, NULL tri-state handling

Q0
7

Distribution of Artworks Across Canvas Size Categories

Mid

Joins: fct_sales → dim_canvas_size

Purpose

Show how artworks and their size-variants are distributed across canvas size buckets. Validates the CASE bucketing logic in dim_canvas_size and shows the portfolio footprint across size tiers. Good smoke test of the size_category logic.

Output Columns

Column	Description
size_category	Small / Medium / Large / Extra Large / Unknown
label_example	One representative label from that bucket
size_id_count	Distinct size_ids in the category
sku_count / distinct_works	Volume metrics
min/max/avg_area_sq_inches	Canvas area range in the bucket
avg_sale_price	Mean sale price
pct_of_total_skus	Share of all SKU rows

Validates: size_category CASE logic, area_sq_inches formula, size_id FK integrity

Q0
8

Top 10 Artists by Total Sale Revenue

Mid

Joins: fct_sales → dim_artist

Purpose

Rank artists by their commercial footprint (total sale revenue across all artworks and size variants). Includes era and nationality context to spot patterns in top earners. Uses DENSE_RANK and LEAD window functions to add relative comparison context.

Output Columns

Column	Description
revenue_rank	DENSE_RANK by total_sale_revenue
artist_id / artist_name	Artist identifier and name
nationality / era / artist_status	Contextual dimension attributes
artwork_count / sku_count	Commercial footprint size
total_sale_revenue / total_regular_revenue	Revenue measures
total_discount_given / avg_discount_pct	Discount metrics

revenue_vs_next_pct	% revenue drop to the next-ranked artist via LEAD
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Validates: *artist FK joins, DENSE_RANK, LEAD window function, era and status attributes*

Q0
9

fct_sales Grain & Surrogate Key Audit

High

Joins: fct_sales only (single table)

Purpose

Verify the deduplication logic in fct_sales is working correctly. The model uses ROW_NUMBER() partitioned by (work_id, size_id) and a surrogate key from dbt_utils.generate_surrogate_key. This audit confirms the grain is clean, all FKs are non-null, and derived columns are accurate.

Output Columns

Column	Description
duplicate_work_size_pairs	Count of duplicate (work_id, size_id) — expected 0
duplicate_sales_keys	Count of duplicate surrogate keys — expected 0
null_work_id / null_size_id	NULL FK counts — expected 0
null_artist_id / null_museum_id	NULL FK counts
discount_calc_mismatches	Rows where stored $\neq$ recomputed — expected 0
total_fact_rows	Total row count for reference

*Validates: ROW_NUMBER dedup, surrogate key uniqueness, FK nullability, discount derivation*Q1
0

Canvas Size Bucket Boundary & Area Computation Audit

High

Joins: fct_sales $\rightarrow$ dim_canvas_size

Purpose

Stress-test the CASE bucketing logic in dim_canvas_size by recomputing area_sq_inches and size_category inline and comparing against stored values. Returns only mismatch rows by default. Validates NULL dimension rows correctly land in 'Unknown'.

Output Columns

Column	Description
size_id / label	Canvas size identifier and label
width_inches / height_inches	Physical dimensions
area_sq_inches	Stored value from the model
expected_area	Recomputed inline (width $\times$ height rounded to 2dp)
area_mismatch	1 if computed $\neq$ stored (>0.01 tolerance)
size_category / expected_category	Stored vs recomputed bucket
category_mismatch	1 if stored $\neq$ expected
sku_count	Rows in fct_sales using this size_id

Validates: *area_sq_inches ROUND formula, size_category CASE boundaries, NULL handling*

Q1 1

Historical vs Living Artist: Pricing & Discount Deep-Dive

Mid-High

Joins: fct_sales → dim_artist

Purpose

Compare pricing dynamics between artists flagged as 'Historical' (death_year IS NOT NULL) and 'Living / Unknown'. Historical works often command premiums. Uses PERCENTILE_CONT for median pricing and compares discount behaviour across the status split.

Output Columns

Column	Description
artist_status	Historical / Living / Unknown
era	Within each status, broken down by era
artist_count / artwork_count / sku_count	Volume metrics
avg_sale_price / median_sale_price	Mean and median using PERCENTILE_CONT
avg_regular_price / avg_discount_pct	Pricing metrics
pct_zero_discount	Fraction of SKUs with 0% or NULL discount
price_premium_ratio	avg_sale_price / overall avg (>1 = premium)

Validates: artist_status CASE logic, PERCENTILE_CONT, CROSS JOIN scalar avg, era × status combinations

Q1 2

City-Level Museum & Artwork Analysis

Mid

Joins: fct_sales → dim_museum

Purpose

Aggregate museum metrics at the city level to identify which cities concentrate the most art supply and what their overall pricing and discount profile looks like. Includes a within-country revenue rank using a window function.

Output Columns

Column	Description
country / city	Location grouping
museum_count	Distinct museums in city
total_artworks / total_sku_rows	Volume metrics
total_sale_revenue / avg_sale_price	Revenue metrics
avg_discount_pct	Mean discount %
avg_opening_days / pct_weekend_museums	Operating profile

city_revenue_rank	RANK within country by total_sale_revenue
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Validates: *dim_museum city/country COALESCE, RANK PARTITION BY country, is_open_weekends aggregation*

Q1
3

Discount Derivation & discount_pct Accuracy Audit

High

Joins: fct_sales only (single table)

Purpose

Precisely audit both derived discount measures in fct_sales: discount_amount and discount_pct. Also checks for business-logic anomalies like negative discounts (sale_price > regular_price) and orphan NULL pct rows where discount_amount ≠ 0.

Output Columns

Column	Description
total_rows	Total fact rows
discount_amount_mismatches	Stored vs recomputed differ > 0.02 — expected 0
discount_pct_mismatches	Stored vs recomputed differ > 0.02 — expected 0
negative_discount_rows	sale_price > regular_price — expected 0
orphan_pct_null_rows	discount_amount ≠ 0 but pct IS NULL — expected 0
zero_price_rows	regular_price = 0 or NULL (pct undefined by design)
rows_with_no_discount	discount_pct = 0 or NULL

Validates: discount_amount ROUND formula, discount_pct formula, NULLIF(regular_price, 0) guard

Q1
4

Multi-Size Artworks: Price Spread & Upsell Opportunity

High

Joins: fct_sales → dim_artwork → dim_canvas_size → dim_artist

Purpose

For artworks available in 3+ size variants, analyse the price spread between their smallest and largest canvas option. A high spread implies strong upsell potential. Also validates that the fct_sales ROW_NUMBER dedup left the correct latest record per (work_id, size_id).

Output Columns

Column	Description
size_variant_count	Distinct size_ids (filtered to >= 3)
work_id / artwork_name / style	Artwork attributes
artist_name / era / nationality	Artist context
min_sale_price / max_sale_price	Cheapest and most expensive size
price_spread	max - min sale price
price_spread_ratio	max / min (upsell multiplier)
size_categories_present	Comma-list of distinct size categories

avg_discount_pct	Mean discount across all sizes
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Validates: Multi-table join chain, STRING_AGG, HAVING on COUNT DISTINCT, dedup correctness

Q1
5

COALESCE / NULL-Fallback Coverage Audit (All Dimensions)

High

Joins: dim_artist, dim_artwork, dim_canvas_size, dim_museum (no fact needed)

Purpose

Every dimension model uses COALESCE or NULLIF to handle missing data. This query counts how many rows actually triggered each fallback so you can quantify data-quality gaps in the source. One summary row per dimension-column combination.

Output Columns

Column	Description
dimension_name	Which model (e.g. dim_artist)
column_name	Which column had the fallback applied
total_rows	Total rows in that dimension
fallback_rows	Rows where the default value is showing
fallback_pct	Percentage of rows using the fallback
fallback_value	The default that was applied (e.g. 'Unknown')

Validates: COALESCE on nationality, style (artist), subject_tags, style NULLIF (artwork), label, size_category (canvas), city, country (museum)

Q1
6

Revenue Impact of Artworks at Canvas Bucket Boundaries

High

Joins: fct_sales → dim_canvas_size

Purpose

Artworks sitting just inside or outside a size-category boundary may have very different price points despite near-identical physical size. Inspects a ±5% band around the 400, 1600, and 4000 sq-inch boundaries and measures the price jump between adjacent buckets.

Output Columns

Column	Description
boundary_label	Human-readable boundary description (e.g. 'Small Medium @ 400')
category_below / category_above	Adjacent bucket names
skus_below / skus_above	Fact rows in the lower/upper bucket within the band
avg_price_below / avg_price_above	Mean sale prices on each side
price_jump_pct	% price increase crossing the boundary

Validates: BETWEEN boundary logic, CASE bucketing edge cases, area_sq_inches accuracy near boundaries

Q1
7

Missing Metadata Orphan Analysis Across All Dimensions

High

Joins: fct_sales → dim_artwork → dim_artist → dim_canvas_size → dim_museum

Purpose

Find fact rows that have degraded joins — mapping to dimension rows that fell into fallback/unknown values. Scores each fact row 0–5 by how many dimensions have unknown metadata and quantifies the revenue at risk from poor data quality.

Output Columns

Column	Description
metadata_quality_score	0 = all clean, up to 5 = all dims unknown
sku_count	Fact rows with that score
distinct_artworks	Distinct work_ids at that quality level
total_sale_revenue / pct_of_total_revenue	Revenue at risk
most_common_flag_combo	Most frequent combination of failing dimensions (MODE)

Validates: Full star-schema join health, CONCAT_WS pipe-flags, MODE aggregate, all COALESCE fallbacks simultaneously

Q1
8

Museum Operating Hours vs Artwork Richness Correlation

Mid-High

Joins: fct_sales → dim_museum

Purpose

Test whether museums with longer opening hours tend to hold more artworks and command higher sale prices. Uses bucketed open-hours ranges (< 6 hrs, 6–8 hrs, 8–10 hrs, 10+ hrs) for BI-friendly grouping. Validates the avg_daily_open_hours derived column.

Output Columns

Column	Description
hours_bucket	Daily open-hours range
museum_count / avg_daily_open_hours	Museum operating profile
avg_opening_days_per_week / pct_weekend_museums	Availability metrics
total_artworks / avg_artworks_per_museum	Artwork richness
avg_sale_price / avg_discount_pct	Pricing metrics
total_revenue	Sum of sale_price in that hours tier

Validates: avg_daily_open_hours EXTRACT(EPOCH) derivation, COALESCE(0) for missing hours, opening_days_per_week

Q1
9

is_in_museum Flag Integrity Audit

High

Joins: fct_sales + dim_artwork (cross-check)

Purpose

The is_in_museum flag in fct_sales is derived as TRUE when museum_id IS NOT NULL (from the artwork join). This audit verifies the flag always agrees with the FK value, and that fct_sales.museum_id aligns with dim_artwork.museum_id for every work_id.

Output Columns

Column	Description
flag_true_but_null_museum	is_in_museum=TRUE yet museum_id IS NULL — expected 0
flag_false_but_has_museum	is_in_museum=FALSE yet museum_id IS NOT NULL — expected 0
fact_museum_vs_artwork_mismatch	fct_sales.museum_id ≠ dim_artwork.museum_id — expected 0
total_in_museum / total_not_in_museum	Distribution of the flag
distinct_museums_in_fact	Distinct non-null museum_ids for reference

Validates: is_in_museum CASE derivation, museum_id propagation from dim_artwork join, FK consistency

Q2
0

Full Star-Schema Stress Test — Revenue Ranking with Percentiles

High

Joins: fct_sales → dim_artist → dim_artwork → dim_canvas_size → dim_museum

Purpose

The definitive end-to-end test. For every (artist_era, size_category, country) combination with at least 5 fact rows, compute revenue metrics, PERCENT_RANK within era, running revenue totals, and a revenue tier classification. If this returns correct results with no duplicates and sensible window values, your entire model is solid.

Output Columns

Column	Description
artist_era / size_category / country	Three-way grouping across all dimension chains
sku_count / distinct_artworks / distinct_artists / museum_count	Volume metrics
total_sale_revenue / avg_sale_price / avg_discount_pct	Revenue metrics
revenue_rank_overall	DENSE_RANK across all combinations
revenue_pct_rank_in_era	PERCENT_RANK within the same artist_era (0–1)
running_revenue_by_era	Cumulative SUM within era ordered by revenue DESC
revenue_tier	'Top 20%' / 'Mid 60%' / 'Tail 20%' from pct_rank

Validates: All 4 dimension joins simultaneously, DENSE_RANK, PERCENT_RANK, cumulative SUM window, HAVING filter

Quick Reference Summary

#	Question	Joins Involved	Difficulty	Validates
Q01	Average Discount % per Artist Era	fct_sales → dim_artist	Mid	era CASE logic in dim_artist, discount_pct accuracy in fct_sales
Q02	Canvas Size Category vs Total Revenue & Avg Discount	fct_sales → dim_canvas_size	Mid	size_category bucketing, area_sq_inches computation, fct_sales → dim_canvas_size...
Q03	Museums with Most Artworks vs Operating Hours	fct_sales → dim_museum	Mid	dim_museum hours enrichment CTEs, COALESCE on city/country, LEFT JOIN to hours_s...
Q04	Share of Artworks Priced Above Median per Nationality	fct_sales → dim_artist	High	PERCENTILE_CONT, CROSS JOIN scalar CTE, per-work price aggregation before nation...
Q05	Artworks with 'Unknown' Subject Tags That Are Discounted	fct_sales → dim_artwork	Mid	COALESCE('Unknown') fallback in dim_artwork.subject_tags, discount_pct accuracy
Q06	Weekend-Open vs Weekday-Only Museum Pricing Comparison	fct_sales → dim_museum	Mid	is_open_weekends boolean derivation in dim_museum, NULL tri-state handling
Q07	Distribution of Artworks Across Canvas Size Categories	fct_sales → dim_canvas_size	Mid	size_category CASE logic, area_sq_inches formula, size_id FK integrity
Q08	Top 10 Artists by Total Sale Revenue	fct_sales → dim_artist	Mid	artist FK joins, DENSE_RANK, LEAD window function, era and status attributes
Q09	fct_sales Grain & Surrogate Key Audit	fct_sales only (single table)	High	ROW_NUMBER dedup, surrogate key uniqueness, FK nullability, discount derivation
Q10	Canvas Size Bucket Boundary & Area Computation Audit	fct_sales → dim_canvas_size	High	area_sq_inches ROUND formula, size_category CASE boundaries, NULL handling
Q11	Historical vs Living Artist: Pricing & Discount Deep-Dive	fct_sales → dim_artist	Mid-High	artist_status CASE logic, PERCENTILE_CONT, CROSS JOIN scalar avg, era × status c...
Q12	City-Level Museum & Artwork Analysis	fct_sales → dim_museum	Mid	dim_museum city/country COALESCE, RANK PARTITION BY country, is_open_weekends ag...
Q13	Discount Derivation & discount_pct Accuracy Audit	fct_sales only (single table)	High	discount_amount ROUND formula, discount_pct formula, NULLIF(regular_price, 0) gu...
Q14	Multi-Size Artworks: Price Spread & Upsell Opportunity	fct_sales → dim_artwork → dim_canvas_size → dim_artist	High	Multi-table join chain, STRING_AGG, HAVING on COUNT DISTINCT, dedup correctness
Q15	COALESCE / NULL-Fallback Coverage Audit (All Dimensions)	dim_artist, dim_artwork, dim_canvas_size, dim_museum (no fact needed)	High	COALESCE on nationality, style (artist), subject_tags, style NULLIF (artwork), l...
Q16	Revenue Impact of Artworks at Canvas Bucket Boundaries	fct_sales → dim_canvas_size	High	BETWEEN boundary logic, CASE bucketing edge cases, area_sq_inches accuracy near ...
Q17	Missing Metadata Orphan Analysis Across All Dimensions	fct_sales → dim_artwork → dim_artist → dim_canvas_size → dim_museum	High	Full star-schema join health, CONCAT_WS pipe-flags, MODE aggregate, all COALESCE...
Q18	Museum Operating Hours vs Artwork Richness Correlation	fct_sales → dim_museum	Mid-High	avg_daily_open_hours EXTRACT(EPOCH) derivation, COALESCE(0) for missing hours, o...
Q19	is_in_museum Flag Integrity Audit	fct_sales + dim_artwork (cross-check)	High	is_in_museum CASE derivation, museum_id propagation from dim_artwork join, FK co...
Q20	Full Star-Schema Stress Test — Revenue Ranking with Percentiles	fct_sales → dim_artist → dim_artwork → dim_canvas_size → dim_museum	High	All 4 dimension joins simultaneously, DENSE_RANK, PERCENT_RANK, cumulative SUM w...